
Clinical Graph-JEPA: Predictive Patient-State Knowledge Graphs for Cognitive Decision Support

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Abstract

Clinical records contain rich evidence about patient state, but converting that evidence into reliable, structured knowledge graphs remains difficult because extraction errors, ontology mismatch, missing relations, and temporal ambiguity can propagate into downstream systems. We propose a clinical knowledge graph construction and refinement framework that combines multi-agent relation proposal, ontology-aware normalization, deterministic evidence scoring, and JEPA-based latent refinement. Rather than treating a clinical knowledge graph as a static extraction artifact, we treat it as a predictive patient-state representation. For each admission, the system constructs an evidence-scored graph from structured MIMIC-IV records and inferred clinical cross-links, then learns to recover held-out clinical relations from the observed graph context. We evaluate the refiner with leakage-free leave-one-out edge recovery (MRR and Hits@k) and held-out batch-mask evaluation (AUC and MRR). To isolate the contribution of discharge-note context, we compare a note-embedding-free configuration with a note-augmented configuration that injects real discharge-note representations only into note-grounded entities. Under the same cohort and evaluation protocol, entity-grounded note injection improves overall leave-one-out MRR by 31% relative improvement.

1 Introduction

Structured clinical data (diagnosis, medication, and procedure tables) records *what* happened to a patient but not the clinician’s *reasoning*: that a β -blocker was started *for* the patient’s atrial fibrillation, that a presenting chest pain *indicated* the acute coronary syndrome that was ultimately worked up. That reasoning lives in the free-text note. A large language model (LLM) can extract it as graph edges, but those inferred edges are exactly the ones with no table-level provenance, so a drafted clinical KG is a mixture of certain structural facts and uncertain inferred relations of widely varying quality. Accordingly, the initial LLM-generated clinical knowledge graph should be treated as an evidence-scored draft rather than a final representation of patient state.

We take the position that the right way to obtain a reliable clinical knowledge graph is not a one-shot extractor but a *refiner*: a model that, given an already-drafted, evidence-scored graph, predicts which edges belong. We instantiate this as a **Clinical Graph-JEPA world model** — a self-supervised graph encoder trained by masked-latent prediction, with a lightweight readout that recovers held-out edges. The model supports two input configurations. Option A is *note-free* and relies only on the consolidated evidence-scored graph, while Option B augments the same graph with Clinical-ModernBERT representations localized to the entities grounded by the discharge note. This distinction preserves a note-independent deployment path while allowing us to quantify the contribution of note context. The results further show that injection strategy is important: a diffuse admission-level note

vector slightly reduces performance, whereas localizing the same representation to note-grounded entities substantially improves inferred-edge recovery. [3]

Contributions. (1) A consolidated, evidence-scored clinical-graph construction that fuses a deterministic MIMIC-IV backbone with LLM-inferred relations and a 14-signal per-edge score. (2) A Clinical Graph-JEPA world model that refines these drafts, evaluated by a leakage-free leave-one-out edge-recovery protocol. (3) Two deployment options — **Option A** (note-free) and **Option B** (note-augmented). (4) The empirical finding that real discharge-note information is most useful when injected *locally* into the entities it grounds, improving Option B over Option A from 0.364 to 0.477 overall leave-one-out MRR.

2 Related Work

Clinical KG extraction. Prior clinical KG systems commonly decompose graph construction into entity identification, entity normalization, relation extraction, and schema or evidence-based validation [4]. Recent LLM-based and multi-agent approaches further specialize these stages to improve coverage and reduce confusion between entity recognition and relation reasoning. However, these methods generally treat the extracted graph as the final representation consumed by downstream systems. Such a graph may combine directly observed structural facts with uncertain text-inferred relations that are incomplete, weakly supported, or incorrectly linked. Our work separates graph construction from graph refinement: the initial graph is treated as an *evidence-scored draft*, and a self-supervised Graph-JEPA refiner learns to recover plausible held-out relations from the surrounding graph context.

Self-supervised and world models. Joint-embedding predictive architectures (JEPA) [1, 6] and BYOL-style latent prediction learn representations by predicting masked latents under an exponential-moving-average (EMA) target with stop-gradient, avoiding contrastive collapse without negatives [2]. We adapt this objective to graphs as the pre-training stage of our refiner, yielding a single shared encoder we treat as a world model over clinical state.

Knowledge-graph completion. Bilinear scoring functions such as DistMult [12] provide an efficient edge-recovery readout. We apply one over the *frozen* world-model encoder, trained with an InfoNCE [10] objective and type-matched negative sampling, so that recovery quality reflects the pre-trained representation rather than a jointly-tuned decoder.

Clinical language models. Domain-pretrained bidirectional encoders (e.g. ModernBERT [11] and its clinical variant, Clinical-ModernBERT [7]) provide note representations. Our contribution is not the note encoder but *where* its output is injected into the graph.

3 Method: Clinical Graphs as Predictive World Models

We formulate clinical knowledge graph refinement as a world-modeling problem over structured patient-state memory. The observation is a structured MIMIC-IV [5] admission record together with its source-faithful clinical narrative, the state is a typed clinical knowledge graph, and revision actions correspond to retaining, reviewing, pruning, or proposing relations. The graph acts as an externalized patient-state representation that organizes diagnoses, medications, procedures, microbiology, and admission context into an auditable structure. Our system does not make autonomous clinical decisions; it improves the structured representation used by downstream retrieval and decision-support systems.

Figure 1 summarizes the pipeline. Panel A shows how structured MIMIC-IV records are converted into patient-state graphs, combining deterministic table-derived edges with narrative-mediated inferred clinical cross-links. Panel B shows Graph-JEPA training, where node features combine entity text embeddings with localized note context, followed by masked patch pretraining and schema-aware graph-revision fine-tuning. Panel C shows inference-time revision, where learned relation plausibility and patch-level predictive consistency are combined before a schema guard assigns the final revision action.

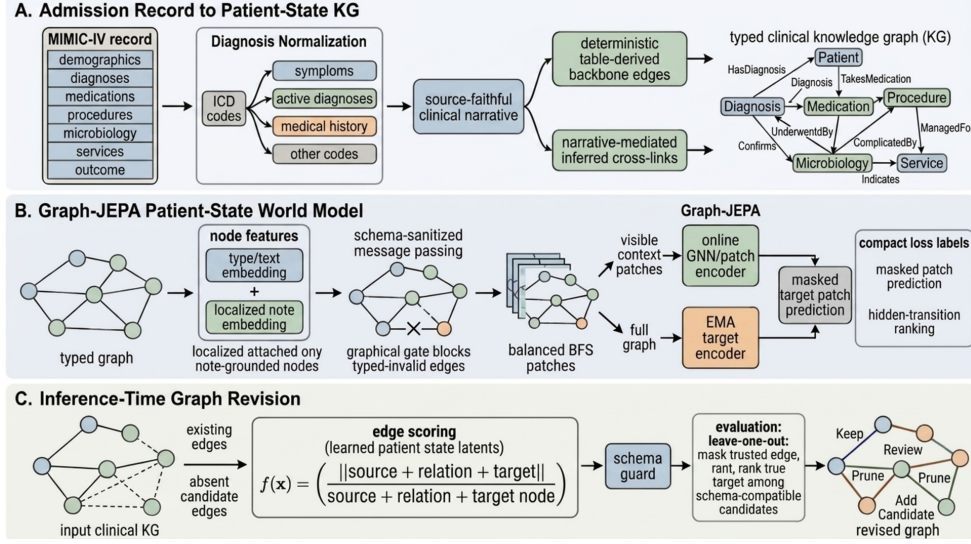


Figure 1: Overview of Clinical Graph-JEPA. (A) Staged extraction converts MIMIC-IV record into evidence-grounded patient-state knowledge graphs. (B) Graph-JEPA learns representations through schema-aware message passing and masked BFS-patch prediction. (C) Inference combines learned edge scoring with schema validation for graph revision, while LOO evaluation measures trusted-edge recovery.

3.1 Consolidated clinical graph

Each admission becomes one *consolidated knowledge graph* fusing two evidence streams. A *deterministic backbone* is read from the MIMIC-IV structured tables, patient → diagnoses, medications, procedures, microbiology, and services (HAS_DIAGNOSIS, TAKES_MEDICATION, UNDERWENT_PROCEDURE, etc) each at confidence 1.0. Over this scaffold, four specialist agents propose the inferred relations the tables do not encode: MANAGED_FOR (medication → diagnosis), CONFIRMS (procedure → diagnosis), COMPLICATED_BY (diagnosis → diagnosis), and INDICATES (presenting symptom → diagnosis).

Deterministic evidence scoring. Every inferred edge receives a deterministic support vector computed from model confidence, biomedical knowledge bases, ontology proximity, and note provenance. These signals include RxNorm linkage, RxNav/MED-RT may-treat evidence, Hetionet [4] treatment and disease-similarity relations, OMOP concept similarity and ancestor distance, and note-grounding checks (prov_in_note, prov_ratio). The same evidence-scored graph is used for both deployment options; the only difference between options is whether note features are provided to the refiner.

3.2 Clinical Graph-JEPA world model

We adapt the JEPA principle to typed clinical knowledge graphs using a masked-node objective, rather than directly following the original Graph-JEPA subgraph-prediction pipeline [9]. The refiner trains in two phases. *Phase 1* is self-supervised masked-latent prediction: a context encoder embeds a masked graph and predicts the latents assigned by an EMA target encoder to the hidden state. The encoder is a typed graph transformer (TransformerConv [8]) over the consolidated graph. Node inputs include a type embedding, a hashed-entity embedding, projected numeric features, and, only under Option B, a localized note vector. *Phase 2* freezes the encoder and trains an edge-recovery readout with an InfoNCE objective using $K=8$ type-matched negatives. This trains the world model to recover graph-state relations from the surrounding patient-state memory rather than from labels or clinical outcomes. Additional details on the entity-localized encoder, masked-node JEPA objective, frozen edge recovery, and leave-one-out evaluation are provided in Appendix A.7–A.12.

3.3 Deployment Options A and B

We evaluate two configurations to isolate the contribution of localized discharge-note representations.

Option A — note-free graph refiner. Option A consumes the consolidated clinical graph without receiving the discharge note or its embedding as a node feature. Its inputs include node type and text features, demographic attributes, typed relations, schema information, edge confidence, and deterministic edge-support metadata. Schema-aware message passing prevents typed-invalid edges from influencing node representations. Graph-JEPA is pretrained by masking connected BFS patches and predicting their latent representations from the visible graph context using an online graph encoder and an EMA target encoder. During revision training, schema-valid trusted edges are treated as positive graph-state transitions, while type-compatible corruptions, typed-invalid observed edges, invalid reversed triples, and weak low-confidence edges provide negatives. A hidden-edge ranking objective removes trusted edges from message passing and trains the model to rank the true target above hard schema-compatible alternatives. Because it requires no note at inference time, Option A applies uniformly to all admissions.

Option B — note-augmented refiner. Option B keeps the same graph, schema, encoder, training losses, and evaluation protocol as Option A, but augments node features when a discharge note is available. The note is encoded once using Clinical-ModernBERT (Simonlee711/Clinical_ModernBERT; 768-dimensional mean-pooled embedding) and injected locally onto note-grounded entities. Nodes not grounded in the note receive a zero note vector. This entity-grounded placement is essential. A single admission-global note vector degrades recovery because it is diffuse and largely redundant with the graph structure extracted from the note. Localizing the same note representation onto the entities it discusses instead provides targeted context for relation revision.

4 Results

4.1 Data and Cohort Construction

We use a fixed random seed sample of MIMIC-IV, a de-identified critical-care EHR [5], comprising 3,018 unique patients, producing 4,000 admission-level patient graphs. The cohort includes patients with single and multiple admissions, with a maximum of 7 admissions per patient. Each graph is paired with a discharge note and its 768-dimensional Clinical-ModernBERT embedding. We evaluate two configurations (Section 3.3) on the same seed-fixed 80/10/10 graph-level split of 3,200 training, 400 validation, and 400 test graphs. This controlled design isolates the contribution of localized discharge-note embeddings under an otherwise identical training and evaluation protocol.

4.2 Limitations

This study has three main limitations. First, all graphs are derived from MIMIC-IV and reflect a single hospital system, so the findings may not generalize to other institutions, populations, coding practices, or clinical workflows. Second, the leave-one-out evaluation measures recovery of held-out relations within an evidence-scored draft graph rather than correctness against an independently adjudicated clinical graph. Finally, the study does not evaluate prospective clinical utility, causal validity, or patient outcomes. The model should therefore be viewed as a graph-refinement component for research and decision-support workflows, not as an autonomous clinical decision-maker.

4.3 Protocol

We evaluate by *leave-one-out* (LOO) edge recovery, the fair link-prediction protocol for a refiner: a single edge is removed (with its inverse, to prevent leakage), the rest of the consolidated graph is kept as context, and the model must recover the held-out edge by ranking the true target against same-type candidates under a *filtered* protocol (other true tails of the query excluded). Ranking is deterministic and reproducible. We report mean reciprocal rank (MRR) and Hits@ k overall and per relation, with chance computed from candidate-set size. We additionally report a held-out batch-mask AUC and a *non-obvious* AUC that excludes trivial patient-hub targets. The evaluation is explicitly of a *refiner*: it improves an already-constructed draft and presupposes the consolidated graph as context, not a cold-start predictor.

4.4 Internal results (MIMIC-IV held-out): Option A vs Option B

Table 1 is the central result. **Option A** (note-free) and **Option B** (note-augmented, on noted admissions) are measured on identical data, split, seed, and *training recipe* (4,000 graphs; 3,200/400/400 split; $n=8,283$ LOO edges); they differ only in the note feature, so Δ is the pure effect of the note.

Inferred relation	Option A (note-free)	Option B (note-augmented)	Δ
MANAGED_FOR	0.326 ± 0.048	0.423 ± 0.029	+0.097
INDICATES	0.469 ± 0.051	0.637 ± 0.047	+0.168
COMPLICATED_BY	0.405 ± 0.056	0.527 ± 0.032	+0.122
CONFIRMS	0.395 ± 0.035	0.506 ± 0.055	+0.111
Overall LOO MRR	0.364 ± 0.045	0.477 ± 0.026	+0.113

Table 1: Leave-one-out edge recovery (MRR), Option A vs Option B, on the four inferred relations. Identical recipe; only the note feature differs.

Option B improves *every* inferred relation, with the largest gains on INDICATES (the presenting-symptom→diagnosis link the note narrates most directly, $0.469 \rightarrow 0.637$) and COMPLICATED_BY ($0.405 \rightarrow 0.527$, $\approx 1.3\times$). The smallest gain is on MANAGED_FOR (+0.097), where Option A already recovers comparatively well (0.326). Option A remains capable of recovering held-out edges without note features, although it consistently underperforms the note-augmented Option B. Crucially, Option A stays well above chance on every relation, which makes it a viable standalone configuration and the fallback inside Option B for note-less admissions.

Metric	Option A	Option B
Batch-mask AUC	0.811 ± 0.003	0.805 ± 0.013
Non-obvious AUC	0.712 ± 0.031	0.800 ± 0.009
Batch-mask MRR	0.358 ± 0.009	0.375 ± 0.005
LOO MRR (overall)	0.364 ± 0.045	0.477 ± 0.026
LOO Hits@1	0.182 ± 0.046	0.303 ± 0.033
LOO Hits@10	0.821 ± 0.037	0.917 ± 0.011

Table 2: Global metrics. The note (Option B) helps on every refiner-relevant measure; the hub-saturated batch-mask AUC is not discriminating.

Table 2 reports global metrics. Option B raises overall LOO Hits@1 ($0.182 \rightarrow 0.303$) and Hits@10 ($0.821 \rightarrow 0.917$) and, on held-out batch-mask scoring, the discriminating *non-obvious* AUC ($0.712 \rightarrow 0.80$). The undifferentiated batch-mask AUC is saturated by trivial patient-hub edges and is not the meaningful measure; the non-obvious AUC and the LOO MRR are.

Configuration	Overall LOO MRR
Option A: note-free (no note)	0.364 ± 0.045
Global NOTE node (diffuse note)	0.352 ± 0.024
Option B: entity-grounded note	0.477 ± 0.026

Table 3: Note-placement ablation. The same note vector helps only when localised onto the entities it grounds.

4.5 Ablation: note placement

Isolating the note feature on identical data confirms that where the note is placed is decisive (Table 3). A diffuse, admission-global note vector is worse than no note at all; localising the same vector onto note-grounded entities is what converts it into signal, and is the form used by Option B. More further ablations can be found in Appendix A.13–A.16

4.6 Where does the recovery come from? A context analysis

To attribute Option B’s recovery we isolate, per inferred relation, three context conditions: a *floor* (only the deterministic backbone present), a *cascade* (the backbone plus earlier inferred relations), and the full LOO *ceiling* (all other edges present). Table 4 shows that the deterministic backbone alone already supplies most of the recoverable context (e.g. INDICATES 0.491 at the floor vs. 0.637 at the ceiling), with the remaining headroom coming from the other inferred cross-links. This indicates the model exploits the free, certain structure first and the inferred relations second — consistent with the refiner framing.

Inferred relation	Floor (backbone)	Cascade	Ceiling (LOO)
MANAGED_FOR	0.300 ± 0.007	0.300 ± 0.007	0.423 ± 0.029
CONFIRMS	0.349 ± 0.017	0.386 ± 0.029	0.506 ± 0.055
COMPLICATED_BY	0.374 ± 0.018	0.433 ± 0.009	0.527 ± 0.032
INDICATES	0.491 ± 0.012	0.527 ± 0.015	0.637 ± 0.047

Table 4: Context analysis for Option B (MRR). The deterministic backbone (floor) provides most of the context; inferred cross-links add the remainder.

5 Discussion

Why entity-grounding works. The note and the graph carry overlapping information — the entities were extracted *from* the note — so a global note summary is largely redundant and adds variance. Placing the note on the specific entities it grounds turns it into a localised prior that distinguishes note-discussed entities from purely structural ones, which is exactly the distinction the inferred relations depend on. This is consistent with two further negative results we observed: feeding the per-edge evidence scores directly into the encoder also failed to help, and a separate global NOTE node hurt. Diffuse, graph-level features add little to a structure-first world model; localised, edge- or entity-specific signal is what helps.

Why two options. We report two configurations to separate the contribution of discharge-note representations from the contribution of the graph itself. Option A provides a note-embedding-free baseline, while Option B measures the additional value of localized note context under otherwise identical conditions. This framing supports deployment in settings with or without note access while providing a controlled estimate of the note contribution.

6 Conclusion

Clinical Graph-JEPA treats an LLM-drafted clinical knowledge graph as an evidence-scored representation to be *refined* rather than accepted as final. The refiner learns to recover held-out, schema-valid clinical relations from the surrounding patient-graph context. We compare two configurations on the same 4,000-graph cohort: **Option A** refines the graph without direct discharge-note embeddings, whereas **Option B** injects localized discharge-note representations into note-grounded entities. Under a matched training and evaluation protocol, entity-grounded note features improve Option B over Option A from 0.364 to 0.477 overall leave-one-out MRR on the evaluation subset.

Future work will evaluate the refiner against independently clinician-annotated relations and across external institutions and patient populations. Additional directions include enforcing patient-disjoint and temporally separated evaluation, developing calibrated graded edge-revision policies, evaluating downstream retrieval and decision-support utility.

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A Additional Method Details

A.1 MIMIC Graph Schema

For MIMIC-derived graphs, each admission graph is represented as $G_i = (V_i, E_i)$ with typed nodes and directed typed edges. The node type set is

$$\mathcal{T}_{\text{MIMIC}} = \{\text{PATIENT, DIAGNOSIS, MEDICATION, PROCEDURE, MICROBIOLOGY, SERVICE}\}. \quad (1)$$

When available, the schema can also include laboratory-test nodes, but in the MIMIC-IV extract used here laboratory values are not available and laboratory features are therefore not used as measured result nodes.

The relation set contains deterministic backbone relations and inferred clinical cross-links:

$$\mathcal{R}_{\text{MIMIC}} = \{\text{HAS_DIAGNOSIS}, \text{TAKES_MEDICATION}, \text{UNDERWENT_PROCEDURE}, \text{HAD_MICROBIOLOGY}, \text{MANAGED_BY_SERVICE}, \text{TARGETS_ORGANISM}, \text{MANAGED_FOR}, \text{CONFIRMS}, \text{COMPLICATED_BY}, \text{INDICATES}\}. \quad (2)$$

The deterministic backbone relations connect the patient node to observed entities in the structured record. Examples include

$$\begin{aligned} \text{PATIENT} &\xrightarrow{\text{HAS_DIAGNOSIS}} \text{DIAGNOSIS}, \\ \text{PATIENT} &\xrightarrow{\text{TAKES_MEDICATION}} \text{MEDICATION}, \end{aligned}$$

and

$$\text{PATIENT} \xrightarrow{\text{UNDERWENT_PROCEDURE}} \text{PROCEDURE}.$$

Microbiology susceptibility data provide direct antibiotic–organism evidence, represented as

$$\text{MEDICATION} \xrightarrow{\text{TARGETS_ORGANISM}} \text{MICROBIOLOGY}.$$

The inferred cross-link schema includes clinically meaningful relations not stored directly as MIMIC relational columns:

$$\begin{aligned} \text{MEDICATION} &\xrightarrow{\text{MANAGED_FOR}} \text{DIAGNOSIS}, \\ \text{PROCEDURE} &\xrightarrow{\text{CONFIRMS}} \text{DIAGNOSIS}, \\ \text{DIAGNOSIS} &\xrightarrow{\text{COMPLICATED_BY}} \text{DIAGNOSIS}, \end{aligned}$$

and

$$\text{DIAGNOSIS} \xrightarrow{\text{INDICATES}} \text{DIAGNOSIS}.$$

Presenting symptoms are represented as diagnosis-layer nodes with a presenting attribute, so symptom–diagnosis links use the same node type while preserving the symptom role in node metadata.

A.2 MIMIC Record Normalization

Each graph is built from a flattened MIMIC-IV admission record. The record contains patient and admission identifiers, demographics, admission and discharge metadata, diagnoses, medications, procedures, microbiology, services, and outcome fields. Diagnoses are partitioned into four clinically distinct buckets: presenting symptoms, active diagnoses, medical history, and other/external codes. This partition prevents historical or external-status codes from being treated as active admission diagnoses.

Medications retain dose, unit, route, and count metadata when available. Microbiology records retain specimen, organism, antibiotic, and susceptibility interpretation. Services and transfer-derived location information are retained as admission context. These fields provide the source of graph nodes and the metadata attached to nodes and edges.

A.3 Graph Construction

Given a normalized admission record r_i , graph construction first creates a deterministic node set

$$V_i = V_i^{\text{patient}} \cup V_i^{\text{dx}} \cup V_i^{\text{med}} \cup V_i^{\text{proc}} \cup V_i^{\text{micro}} \cup V_i^{\text{service}}. \quad (3)$$

The deterministic backbone edge set is

$$E_i^{\text{backbone}} = B(r_i), \quad (4)$$

where B adds direct table-derived relations such as diagnoses, medications, procedures, microbiology, services, and susceptibility-supported antibiotic–organism links.

To infer clinical cross-links, the structured record is rendered into a source-faithful synthetic note narratives $n_i = N(r_i)$. The narrative is constrained to mention only facts present in r_i . Relation-specific extraction agents then operate over candidate pairs of existing graph nodes:

$$E_i^{\text{cross}} = A_{\text{managed}}(V_i, n_i) \cup A_{\text{confirms}}(V_i, n_i) \cup A_{\text{complicated}}(V_i, n_i) \cup A_{\text{indicates}}(V_i, n_i). \quad (5)$$

The final graph is

$$G_i = (V_i, E_i^{\text{backbone}} \cup E_i^{\text{cross}}). \quad (6)$$

The extraction agents are constrained to link only nodes already present in V_i . Candidate edges with unmatched endpoints, invalid type signatures, self-loops, duplicate entries, or unsupported evidence are removed during validation.

A.4 Provenance and Edge Support

Each edge stores its source node, target node, relation type, confidence, source field or evidence text, and provenance metadata. Backbone edges are directly grounded in MIMIC tables and are assigned full confidence. Inferred cross-links store cited evidence from the generated narrative and are further annotated with deterministic support signals.

For medication–diagnosis relations, support includes biomedical semantic similarity and treatment evidence from resources such as RxNorm/RxNav and Hetionet [4]. Additional support features include OMOP concept similarity or ontology proximity when available, as well as provenance checks indicating whether the cited evidence appears in the narrative. These scores are retained as continuous edge attributes rather than collapsed into hard-coded rules. This allows downstream models to learn how to use support signals while preserving the original high-recall graph.

A.5 Graph Unit and Entity Identity

The training unit is an admission-level graph. Entities are grounded in the structured MIMIC record before relation extraction, and inferred edges are restricted to those existing entities. Therefore, the current MIMIC pipeline does not require corpus-level embedding-based entity resolution to construct the training graphs. Node normalization is performed within the admission record, and node metadata retains the original MIMIC-derived names, codes, and source fields needed for provenance.

A.6 Node and Note Features

Each node is represented by a base embedding of its type and canonical text:

$$b_v = f_\phi(\tau(v), \text{text}(v)). \quad (7)$$

When an admission-level note embedding n_i is available, it is localized to nodes grounded in the note. Let $a_v \in \{0, 1\}$ indicate whether node v is grounded. The final node feature is

$$x_v = [b_v, a_v n_i]. \quad (8)$$

If no note embedding is available, the note component is zero.

By default, a node is considered grounded when it participates in an edge whose provenance check indicates support in the note. We also support grounding by surface-name match to the note, or assigning the note embedding to all non-patient nodes. This keeps note context localized to clinically grounded entities rather than broadcasting it uniformly to the whole graph.

A.7 GNN Encoder, Graph Patching, and Target Encoding

The Graph-JEPA encoder uses typed message passing over the schema-sanitized message graph. Relation labels are embedded and used as edge attributes in the GNN. The final node latent is

$$z_v = W_o h_v^{(L)}. \quad (9)$$

The graph is partitioned into K local patches using balanced multi-source BFS. Patch representations are mean-pooled node latents:

$$q_j = \frac{1}{|P_j|} \sum_{v \in P_j} z_v. \quad (10)$$

Patch positional features include relative patch size, patch degree, and random-walk return features on the coarsened patch graph.

For each graph, context patches C and target patches T are sampled. The online encoder receives visible context patches and learned mask content for hidden patches. The EMA target encoder receives the full graph and is updated by

$$\bar{\theta} \leftarrow m\bar{\theta} + (1 - m)\theta. \quad (11)$$

For each $j \in T$, the predictor maps the context patch state and positional features to $\hat{q}_j$.

A.8 Hyperparameters

Table 5: Principal hyperparameters used in the pipeline

Stage	Hyperparameter	Value
Graph encoder	Architecture	TransformerConv
	Hidden dimension	128
	Number of layers	2
	Attention heads	4
	Relation embedding dimension	32
JEPA pretraining	Pretraining epochs	60
	Batch size	16
	Learning rate	1×10^{-3}
	Target-node mask ratio	0.40
	EMA coefficient	$0.996 \rightarrow 0.9999$
	Prediction loss	Cosine distance
Edge readout	Readout epochs	40
	Encoder during readout	Frozen
	Decoder	MLP
	Objective	InfoNCE
	Negative samples per positive	8
	Edge-mask ratio	$\mathcal{U}(0.10, 0.60)$
	Target-relation weight	3.0

A.9 Masked-Node JEPA Loss

The self-supervised objective predicts normalized target-encoder latents for masked nodes. Let $\hat{z}_v$ be the predictor output and $\bar{z}_v$ the stop-gradient target latent. The loss is cosine prediction error:

$$\mathcal{L}_{\text{JEPA}} = \frac{1}{|M|} \sum_{v \in M} \left(2 - 2 \frac{\hat{z}_v^\top \bar{z}_v}{\|\hat{z}_v\|_2 \|\bar{z}_v\|_2} \right). \quad (12)$$

A.10 Frozen Edge Recovery

After JEPA pretraining, the graph encoder is frozen and a MLP readout is trained for edge recovery. For a candidate triple $e = (u, r, v)$, the score is

$$s_\phi(u, r, v) = \sum_{\ell=1}^h a_\ell \sigma(\mathbf{w}_\ell^u \cdot \mathbf{Z}_u + \mathbf{w}_\ell^v \cdot \mathbf{Z}_v + \mathbf{w}_\ell^r \cdot \mathbf{Z}_r + b_\ell) + b_0, \quad (13)$$

where $\mathbf{Z}_u$ and $\mathbf{Z}_v$ denote the source and destination node embeddings, respectively, and $\mathbf{Z}_r$ denotes the relation embedding. The vectors $\mathbf{w}_\ell^u$, $\mathbf{w}_\ell^v$, and $\mathbf{w}_\ell^r$ are learned weights for the ℓ -th hidden unit, b_ℓ is its bias, a_ℓ is the learned output-layer weight, b_0 is the output bias, h is the number of hidden units, and $\sigma(\cdot)$ is the ReLU activation function. For each hidden positive edge, negative targets are sampled from nodes of the same type. With candidate set $\mathcal{C}_i = \{e_i\} \cup \mathcal{N}_i$, the ranking loss is

$$\mathcal{L}_{\text{rank}} = -\frac{1}{|H|} \sum_{i \in H} \log \frac{\exp(s_\phi(e_i)/T)}{\sum_{e \in \mathcal{C}_i} \exp(s_\phi(e)/T)}. \quad (14)$$

A.11 Leave-One-Out Evaluation

For leave-one-out edge recovery, one trusted edge (u, r, v) is removed from the message-passing graph while the remaining graph context is retained. The source node u and relation type r are held fixed, and the model ranks the true target node v against candidate target nodes of the same semantic type. Other known true targets for the same source–relation query are filtered from the candidate set.

The graph encoder and the MLP readout are kept fixed during evaluation. Performance is reported using mean reciprocal rank (MRR) and Hits@ k , with $k \in \{1, 3, 10\}$. This protocol measures the recovery of held-out clinical relationships from learned patient-state latents. It does not perform inference-time graph revision, graph completion, or iterative edge insertion.

A.12 Candidate Edge Interpretation

ClinG-JEPA scores source–relation–target triples as recovery candidates. High scores indicate that a candidate edge is plausible under the learned patient-state representation, but they are not treated as verified clinical facts. Unlike the modular graph-revision pipeline, we do not assign explicit Keep, Review, Prune, or Add-Candidate actions.

A.13 External Validation on ACI-Bench

We evaluated the model on the independent ACI-Bench dataset to assess whether the entity-grounded note-injection finding remains valid beyond the MIMIC-based evaluation. The evaluation used all 207 available encounters and was performed without retraining the model on ACI-Bench.

We constructed the knowledge graphs using the same multi-agent pipeline used above for the MIMIC data. The resulting graph records were embedded using the same Clinical-ModernBERT embedding procedure used in the primary pipeline. We evaluated three conditions using the same model checkpoint, graph topology, candidate-generation procedure, and leave-one-out edge-recovery protocol. Option A used no note information, the global-note condition, and Option B injected the note embedding only into entities supported by note provenance. The results are shown in Table 6.

Table 6: External validation on the 207-record ACI-Bench dataset. All conditions were evaluated using the same model checkpoint and the same 5,005 rankable leave-one-out queries.

Condition	MRR	Hits@1	Hits@3	Hits@10
Option A: No note	0.475	0.304	0.561	0.844
Global note injection	0.472	0.301	0.554	0.855
Option B: Entity-grounded note injection	0.481	0.305	0.570	0.865

As shown in Table 6, entity-grounded note injection achieved the highest MRR and the highest Hits@1, Hits@3, and Hits@10 values. Compared with global note injection, entity-grounded injection improved MRR by approximately +0.009, with a 95% confidence interval of [0.006, 0.013]. These results indicate that the entity-grounded injection finding also holds on the independent ACI-Bench dataset, supporting the benefit of localizing note information to the entities it supports rather than applying the same note representation uniformly across the graph.

A.14 Discharge-Note Representation Ablation

We compared three note representations while keeping the same entity-grounded injection strategy: a global mean embedding, uniform pooling of local note spans, and entity-conditioned attention over local spans. The purpose was to determine whether preserving local note information improves graph-edge recovery over the global mean representation. The experiment used the same datasets and configurations described above.

Table 7: Discharge-note representation ablation. Values are mean $\pm$ standard deviation across ten paired seeds. All note representations were injected only into entities with note provenance.

Note representation	LOO MRR	Hits@1	Hits@10
Entity-grounded global mean	0.468 $\pm$ 0.021	0.291 $\pm$ 0.026	0.911 $\pm$ 0.008
Entity-grounded uniform local spans	0.478 $\pm$ 0.020	0.302 $\pm$ 0.027	0.915 $\pm$ 0.005
Entity-conditioned attention	0.458 $\pm$ 0.026	0.280 $\pm$ 0.030	0.911 $\pm$ 0.011

The entity-grounded global-mean condition used one 768-dimensional Clinical ModernBERT embedding for the complete discharge note and copied it to each entity with note provenance. The uniform local-span condition instead represented each grounded entity using a token-count-weighted mean of its available local note spans. The attention condition used an entity-specific query to attend over the available spans.

As shown in Table 7, uniform local-span pooling achieved the highest numerical LOO MRR (0.478), compared with 0.468 for the entity-grounded global mean. However, its paired improvement of +0.010 had a 95% confidence interval of $[-0.013, +0.032]$, which includes zero. Entity-conditioned attention performed worse than both alternatives, with an LOO MRR of 0.458 and a paired difference of -0.020 relative to uniform local-span pooling.

A.15 Note-Encoder Ablation

We evaluated whether entity-grounded note gains depend on the quality of the note encoder. The full graph-learning and edge-recovery pipeline was kept fixed, including the graph, entity-grounding procedure, masking strategy, training schedule, data split, and evaluation protocol. Only the node-level note encoder was changed: TF-IDF embeddings were compared with the existing Clinical-ModernBERT embeddings.

Table 8: Entity-grounded note-encoder ablation. Values are mean $\pm$ standard deviation across seeds.

Metric	Clinical-ModernBERT	TF-IDF
Batch-mask MRR	0.37 $\pm$ 0.004	0.36 $\pm$ 0.003
Filtered LOO MRR	0.47 $\pm$ 0.02	0.38 $\pm$ 0.04
Filtered LOO Hits@1	0.29 $\pm$ 0.03	0.19 $\pm$ 0.04
Filtered LOO Hits@3	0.54 $\pm$ 0.02	0.43 $\pm$ 0.04
Filtered LOO Hits@10	0.91 $\pm$ 0.01	0.84 $\pm$ 0.02

As shown in Table 8, TF-IDF encoder underperformed Clinical-ModernBERT on all primary LOO metrics, indicating that entity grounding alone does not account for the full recovery gain. The result suggests that note-encoder quality also contributes substantially to graph-edge recovery.

A.16 Graph-QA Ablation

We conducted a question-answering ablation to measure the contribution of clinical note information to graph-based reasoning. We created a fixed benchmark of 20 clinical questions and corresponding answers using the GPT-5.6 Luna model and the MIMIC dataset. The benchmark included direct graph-recall questions, clinical metadata questions, and multi-hop questions requiring the composition of multiple graph relations. The same questions, graph contexts, candidate answer sets, and gold answers were evaluated under two configurations described above.

For each question, the model ranked candidate graph entities and produced a predicted answer set. We report macro entity-level F1, question-level Hits@ k , and strict question exact match. Question Hits@ k measures the fraction of questions for which every required answer group contains at least one correct entity within the top- k ranked candidates. Strict question exact match requires the complete predicted answer set for every answer group to exactly match the gold answer set.

Table 9: Option A–B graph question-answering ablation on the fixed 20-question benchmark. Values are percentages.

Metric	Option A	Option B	Δ (B–A)
Macro entity F1	70.6	74.1	+3.6 pp
Question Hits@1	65.0	65.0	+0.0 pp
Question Hits@3	65.0	70.0	+5.0 pp
Strict question exact match	65.0	65.0	+0.0 pp

As shown in Table 9, Across the 20-question benchmark, Option B improved macro entity F1 by 3.6 percentage points and question-level Hits@3 by 5.0 percentage points, while strict question exact-match accuracy remained unchanged at 65.0%.